Lab - Azure Container Instances (ACI)
To Begin with the lab

Summary

This lab demonstrates how to deploy and run a Docker container using **Azure Container Instances (ACI)**. Unlike Azure Virtual Machines, ACI eliminates the need to manage servers or install Docker. You simply select a container image, configure networking, and Azure automatically provisions the infrastructure to run your container.

Understanding

Azure Container Instances (ACI) is a serverless container service that allows you to run Linux or Windows containers without managing virtual machines. Azure automatically provides the compute resources and Docker runtime. You only need to specify the container image, CPU/Memory requirements, networking, and ports.

ACI is best suited for simple containerized applications, testing, development environments, scheduled jobs, and short-lived workloads. For large-scale applications requiring orchestration, services like Azure Kubernetes Service (AKS) or Azure Container Apps are more appropriate.

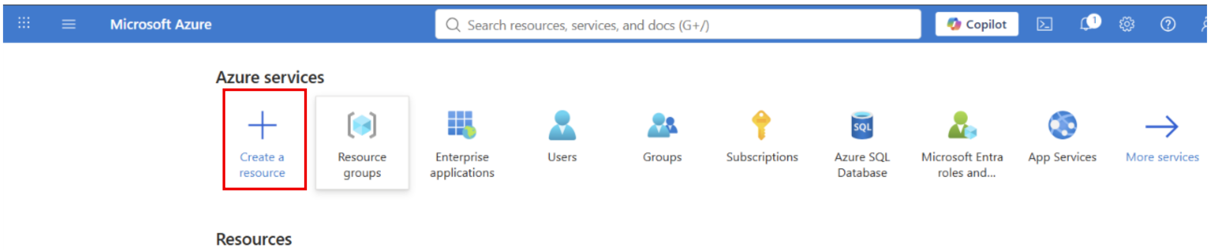
Example:

- Azure VM → You create and manage the VM, install Docker, and run containers.
- Azure Container Instance → Azure manages the infrastructure. You only deploy the container image.

Lab Steps

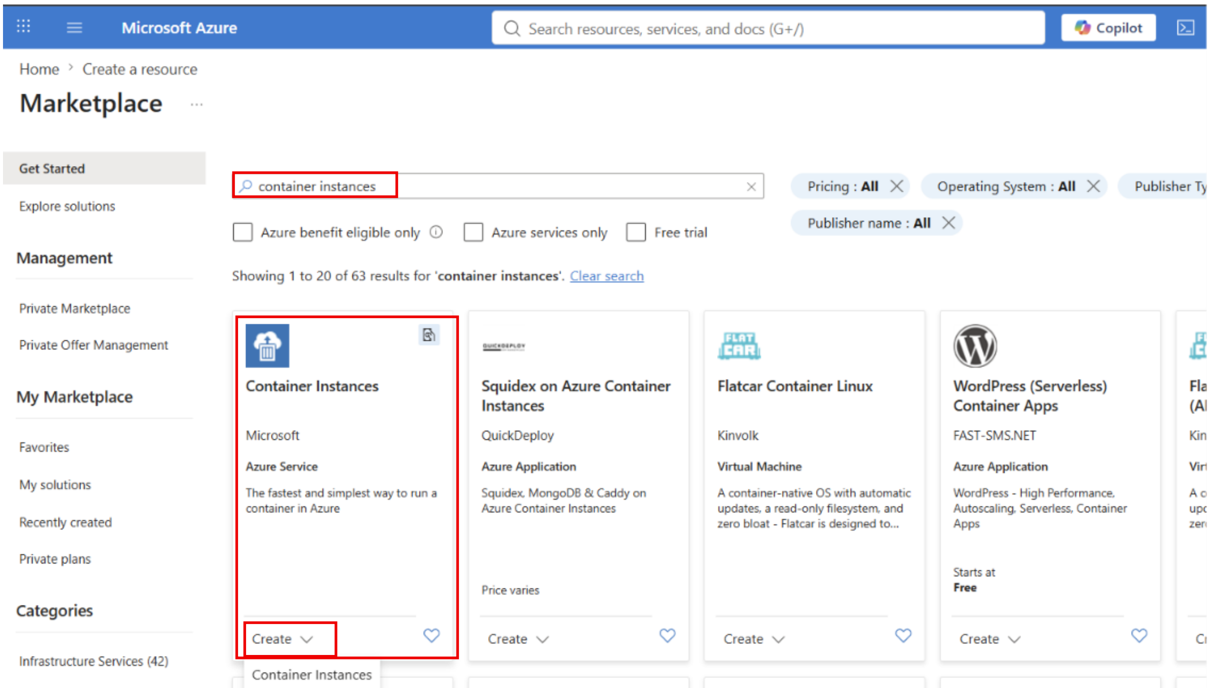
Step 1: Open Azure Portal

- Sign in to the Azure Portal.
- Open **All Resources** (optional).
- Click **Create a Resource**.



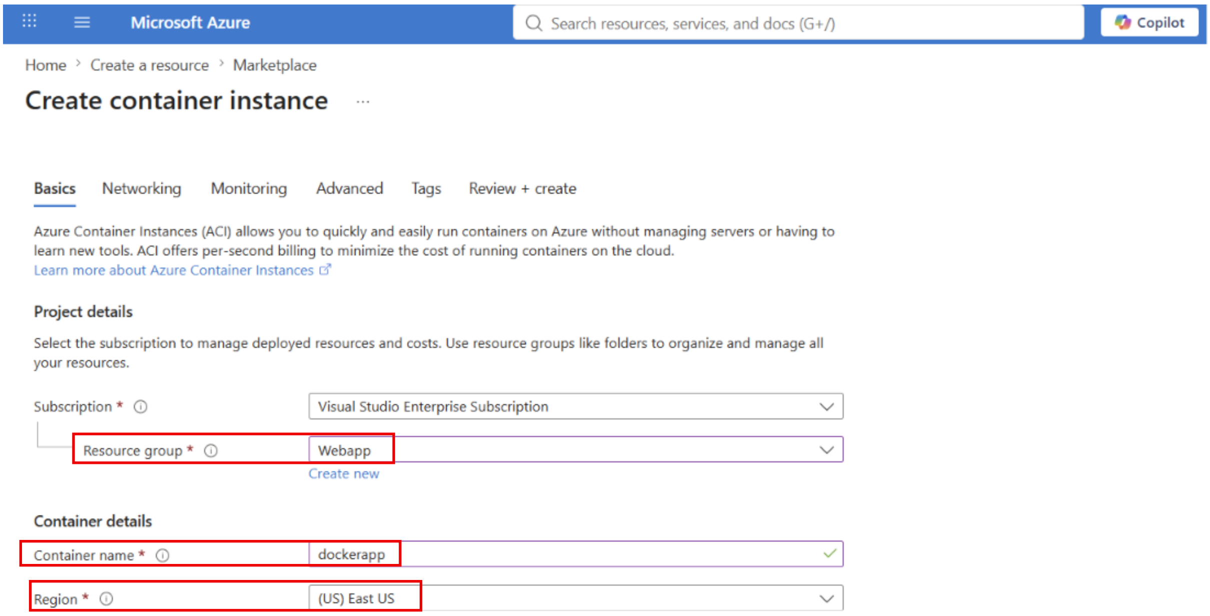
Step 2: Search for Container Instance

- Search for **Container Instances**.
- Select **Azure Container Instances**.
- Click **Create**.



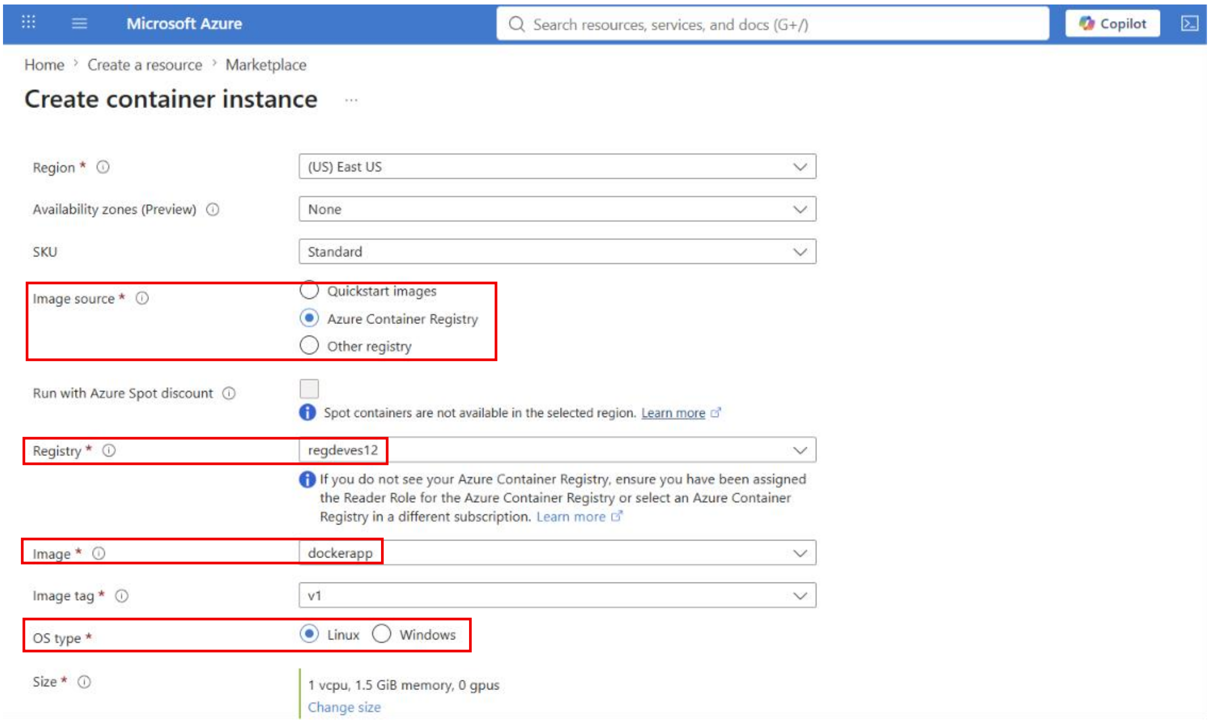
Step 3: Configure Basic Settings

- Select the existing **Resource Group**.
- Enter a **Container Name** (Example: dockerapp).
- Choose the **Region** (Example: East US).



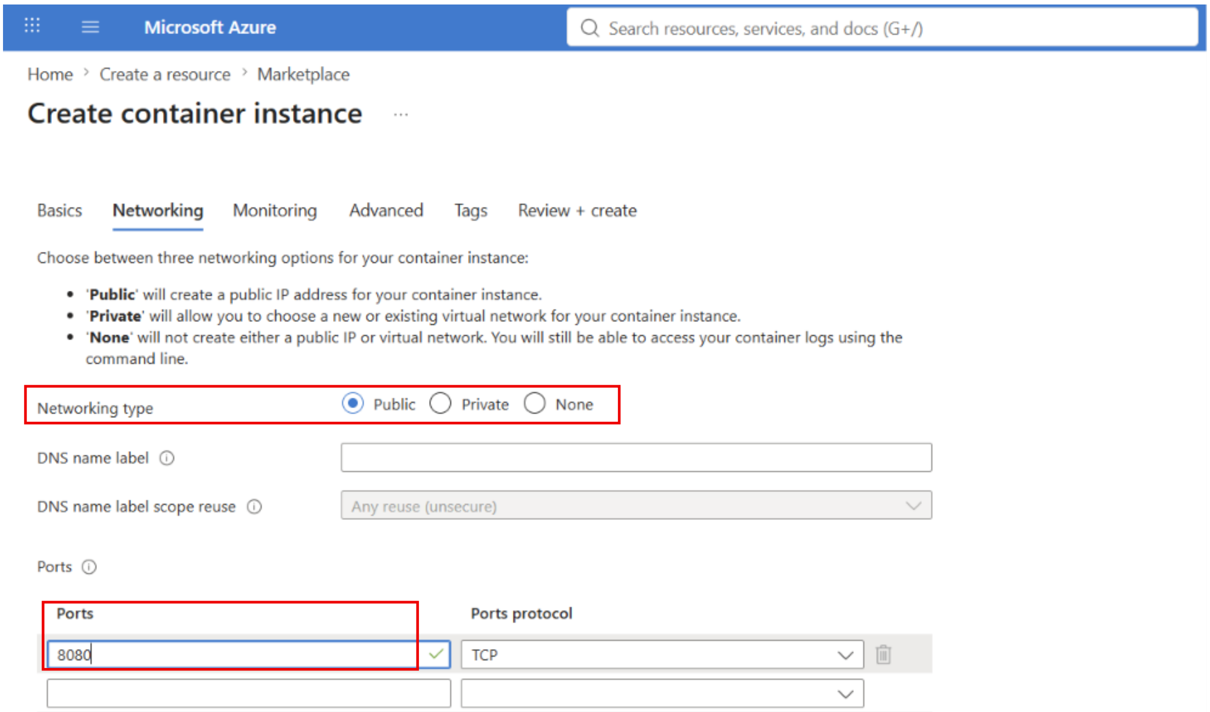
Step 4: Select Container Image

- Under **Image Source**, choose **Azure Container Registry**.
- Select your **Container Registry**.
- Select the Docker **Image**.
- Choose the appropriate **Image Tag**.
- Keep **OS Type** as **Linux**.



Step 6: Configure Networking

- Go to the **Networking** tab.
- Set **Public IP** to **Enabled**.
- Add **Port 8080**.
- Keep protocol as **TCP**.



Step 7: Monitoring

- Open the **Monitoring** tab.
- Leave **Container Insights** with the default settings.

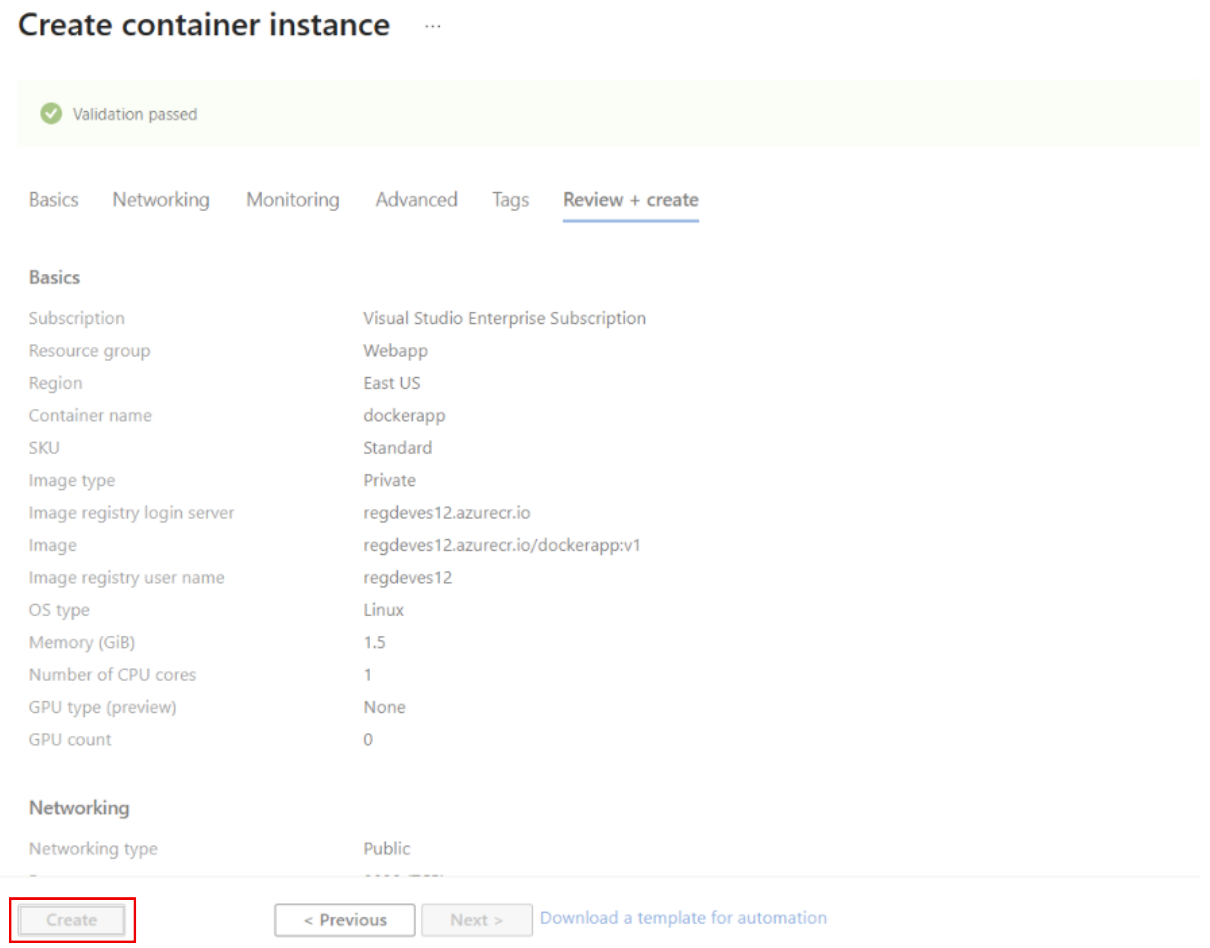
Step 8: Advanced Settings

- Open the **Advanced** tab.

- Leave the default settings unless customization is required.

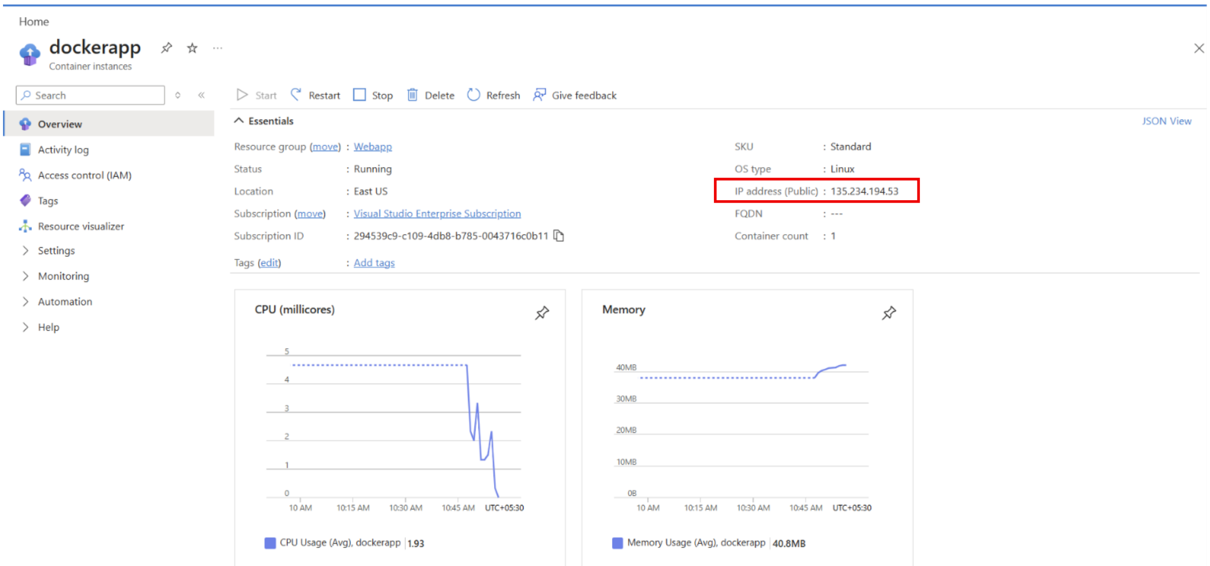
Step 9: Deploy the Container

- Click **Review + Create**.
- Validate the configuration.
- Click **Create**.
- Wait until deployment completes.

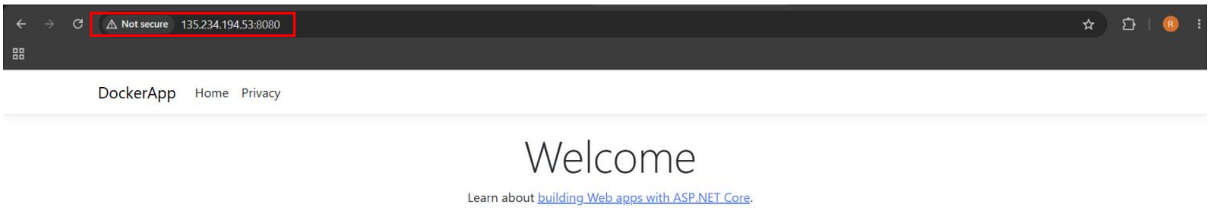


Step 10: Open the Running Application

- Open the deployed **Container Instance**.
- Copy the **Public IP Address**.



- Open a browser.
- Browse to:
- http://<Public-IP>:8080



- Verify that the Docker application is displayed successfully.